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Linux Containers and Kernel Sharing

In simple terms, Linux containers are a lightweight alternative to virtualization.

A virtual machine contains and runs the entire guest operating system in a virtualized environment.

The virtual machine, in turn, runs on top of an environment such as a hypervisor that manages access to the physical resources of the host system.

Container Uses and Advantages

The main advantage of containers is that they require considerably less resource overhead than virtualization allowing many container instances to be run simultaneously on a single server.

They can be started and stopped rapidly and efficiently in response to demand levels.

In addition, containers run natively on the host system providing a level of performance that a virtual machine cannot match.

Containers are also highly portable and can be easily migrated between systems. Combined with a container management system such as Docker, OpenShift, and Kubernetes,

it is possible to deploy and manage containers on a vast scale spanning multiple servers and cloud platforms, potentially running thousands of containers.

Containers are frequently used to create lightweight execution environments for applications.

In this scenario, each container provides an isolated environment containing the application together with all of the runtime and supporting files required by that application to run.

The container can then be deployed to any other compatible host system that supports container execution and runs without any concerns that the target system may not have the necessary runtime configuration for the application - all of the application's dependencies are already in the container.

Working with Containers on CentOS Stream 9

Before starting with containers, the first step is to install all of the container tools using the following command:

```
# dnf install container-tools
```

```
# podman pull quay.io/centos/centos:stream9-minimal
```

```
# podman images
```

```
podman inspect quay.io/centos/centos:stream9-minimal
```

```
root@centos9 ~]# # podman images
```

```
[root@centos9 ~]# podman run --rm quay.io/centos/centos:stream9-minimal cat /etc/passwd
```

```
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
[root@centos9 ~]#
```

```
# podman run --name=mycontainer -it quay.io/centos/centos:stream9-minimal /bin/bash
```

```
[root@centos9 ~]# podman ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED
7fb97fe842ed	docker.io/ollama/ollama:latest	serve	29 hours ago
Exited (0) 29 hours ago	0.0.0.0:11434->11434/tcp	ollama	
a4a7769f1ce2	quay.io/centos/centos:stream9-minimal	/bin/	5 hours ago
Created		mycontainer	
52d8661ab3b0	quay.io/centos/centos:stream9-minimal	/bin/bash	5 hours ago
hours		mycontainer2	Up 5
cf6fdb7b472	localhost/my-coreutils-image:latest	bash	4 hours ago
Exited (127) 4 hours ago		laughing_chatelet	
d4abe80a2e1a	localhost/my-coreutils-image:latest	bash	4 hours ago
Exited (0) 5 minutes ago		epic_jang	

```
podman run --name=mycontainer4 -it quay.io/centos/centos:stream9-minimal /bin/bash
bash-5.1#
```

THE PREVIOUS PROMPT IS FROM THE NEW CONTAINER

IN ANOTHER TERMINAL SESSION

```
[bobb@centos9 ~]$ podman ps -a
```

```
[bobb@centos9 ~]$ podman ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED
7fb97fe842ed	docker.io/ollama/ollama:latest	serve	30 hours ago
Exited (0) 29 hours ago	0.0.0.0:11434->11434/tcp	ollama	

```

a4a7769f1ce2 quay.io/centos/centos:stream9-minimal /bin/ 5 hours ago
Created mycontainer
52d8661ab3b0 quay.io/centos/centos:stream9-minimal /bin/bash 5 hours ago Up
5 hours mycontainer2
cf6fdb7b472 localhost/my-coreutils-image:latest bash 5 hours ago
Exited (127) 5 hours ago laughing_chatelet
d4abe80a2e1a localhost/my-coreutils-image:latest bash 4 hours ago
Exited (0) 23 minutes ago epic_jang
cd257bdb2571 quay.io/centos/centos:stream9-minimal /bin/bash 5 minutes ago
Exited (0) 4 minutes ago mycontainer4
b5a62da4ecff quay.io/centos/centos:stream9-minimal /bin/bash 4 minutes ago Up
4 minutes mycontainer5
[bobb@centos9 ~]$

```

```

[bobb@centos9 ~]$ podman exec -it mycontainer5 /bin/bash
bash-5.1#

```

```

bash-5.1# pwd
/
bash-5.1# exit
exit
[bobb@centos9 ~]$ podman attach mycontainer5

```

```

bash-5.1#

```

```

podman start mycontainer5

```

```

[bobb@centos9 ~]$ podman ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED
STATUS        PORTS                                NAMES
...
b5a62da4ecff   quay.io/centos/centos:stream9-minimal /bin/bash               14 minutes ago Up
3 seconds                                mycontainer5
[bobb@centos9 ~]$

```

```

[bobb@centos9 ~]$ podman exec -it mycontainer5 /bin/bash
bash-5.1#

```

```

IN ANOTHER TERMINAL SESSION

```

```

# podman pause mycontainer5

```

```

podman ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED
STATUS        PORTS                                NAMES
...
b5a62da4ecff   quay.io/centos/centos:stream9-minimal /bin/bash               24 minutes ago
Paused                                mycontainer5

```

```

IN THE CONTAINER IN THE OTHER TERMINAL SESSION YOU CAN'T TYPE A COMMAND

```

```

# podman unpause mycontainer5

```

```

YOU CAN NOW TYPE IN COMMANDS IN THE OTHER TERMINAL SESSION

```


cutting-edge humanoid robot designed for assembly-line work. Robby's creators believed that by giving him advanced sensors and processing power, he could learn to perform complex tasks with precision and speed...

...

>>> /bye

```
[bobb@centos9 ~]$ podman ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
90f10794a50d	docker.io/ollama/ollama:latest	serve	13 minutes ago	Up 13 minutes

0.0.0.0:11434->11434/tcp ollama

```
[bobb@centos9 ~]$ curl http://localhost:11434/api/generate -d '{
"model": "llama3",
"prompt": "Why use containers?",
"stream": false
}'
```

```
{"model":"llama3","created_at":"2026-08-05T20:37:57.949319319Z","response":"Containers, also known as containerization, have become increasingly popular in recent years, and for good reason..."}
...
[bobb@centos9 ~]$
```

```
podman exec -it ollama ollama run llama3
```

It executes a command inside a running Podman container named ollama, opens an interactive shell, and then runs the Llama-3 model through Ollama inside that container.



Breakdown of each part

1. podman exec

Runs a command inside an already-running container.

2. -it

-i → interactive (keeps STDIN open)

-t → allocate a pseudo-TTY

This lets you talk to the model interactively.

3. Container name: ollama

This tells Podman which container to enter.

4. Command inside the container: ollama run llama3

Runs the Llama-3 model using the Ollama runtime installed inside the container.

If the model isn't already pulled, Ollama will download it first.



What actually happens on your system

Podman attaches your terminal to the container.

Inside the container, it launches the Ollama CLI.

Ollama loads the llama3 model.

You get an interactive prompt where you can type messages to the model.

Creating a container from scratch

```
[bobb@centos9 ~]$ buildah from ubuntu:22.04
```

```
ubuntu-working-container-3
```

```
[bobb@centos9 ~]$ buildah mount ubuntu-working-container
```

```
Error: cannot mount using driver overlay in rootless mode. You need to run it in a
`buildah unshare` session
```

```
[bobb@centos9 ~]$ buildah unshare
```

```
[root@centos9 ~]# buildah mount ubuntu-working-container
```

```
/home/bobb/.local/share/containers/storage/overlay/83e3d0982907c745c6ef78d7b16118ce
320ca721865e1efa68d1d5d9e52c152f/merged
```

```
[root@centos9 ~]# buildah run ubuntu-working-container -- apt-get update
```

```
buildah run ubuntu-working-container -- apt-get install -y coreutils
```

```
Hit:1 http://security.ubuntu.com/ubuntu jammy-security InRelease
```

```
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
```

```
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
```

```
Hit:4 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
```

```
Reading package lists... Done
```

```
Reading package lists... Done
```

```
Building dependency tree... Done
```

```
Reading state information... Done
```

```
coreutils is already the newest version (8.32-4.1ubuntu1.3).
```

```
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

```
[root@centos9 ~]# buildah umount ubuntu-working-container
```

```
[root@centos9 ~]# buildah commit ubuntu-working-container my-coreutils-image
```

```
Getting image source signatures
```

```
Copying blob beb59027b96d skipped: already exists
```

```
Copying blob af95ea819889 done    |
Copying config 8b37535a8c done    |
Writing manifest to image destination
8b37535a8ca04000a6c91ed9f75a428399e34f953223af4673d00cd3078acd91
```

```
[root@centos9 ~]# podman run -it my-coreutils-image bash
root@d4abe80a2e1a:/#
```